

DEMO: The GhostTwin: Towards Live Digital Twins via SmartNIC Network Emulation

Francisco Germano Vogt
Universidade Estadual de Campinas

Victor H. S. Lopes
Federal University of Pampa (Unipampa)

Fabricio Rodriguez
Telefonica Research

Marcelo Caggiani Luizelli
Federal University of Pampa (Unipampa)

Chrysa Papagianni
University of Amsterdam (UvA)

Christian Rothenberg
Universidade Estadual de Campinas

ABSTRACT

Network Digital Twin (NDT) technology enables the replication of entire network infrastructures into fully synchronized digital replicas. This capability is especially timely for the safe integration of AI into critical networks, where model correctness and reliability are essential. In this work, we introduce GhostTwin, the first open-source framework for defining and operating high-fidelity NDTs in a self-contained environment. GhostTwin employs SmartNIC-based hardware emulation to realistically replicate programmable physical networks, capturing network performance metrics such as packet loss and residual bandwidth. The framework supports diverse telemetry sources, enabling scalable and accurate digital twin deployments in a programmable testbed.

CCS CONCEPTS

• **Networks** → **Network experimentation; Network simulations; • Hardware** → *Emerging technologies.*

KEYWORDS

Digital Twins, Programmable Networks, Network Emulation.

ACM Reference Format:

Francisco Germano Vogt, Victor H. S. Lopes, Fabricio Rodriguez, Marcelo Caggiani Luizelli, Chrysa Papagianni, and Christian Rothenberg. 2025. DEMO: The GhostTwin: Towards Live Digital Twins via SmartNIC Network Emulation. In *ACM SIGCOMM 2025 Conference (SIGCOMM Posters and Demos '25)*, September 8–11, 2025, Coimbra, Portugal. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3744969.3748450>

1 INTRODUCTION

Modern networks are growing in size, complexity, and demands, making them increasingly difficult to control and optimize [1, 3]. As services require low latency, high reliability, and real-time adaptability, network operators must make infrastructure smarter, more agile, and responsive. Achieving this without compromising performance or stability is a major challenge [7]. Network Digital Twins (NDTs) address this need by maintaining a dynamic, real-time replica of the physical network, allowing operators to simulate outcomes, detect issues proactively, and safely test changes before

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SIGCOMM Posters and Demos '25, September 8–11, 2025, Coimbra, Portugal

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-2026-0/2025/09.
<https://doi.org/10.1145/3744969.3748450>

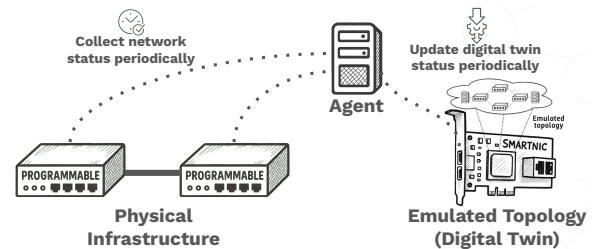


Figure 1: GhostTwin demo overview.

deployment [2, 7]. This is especially valuable when integrating AI into network infrastructure, where model correctness and reliability are critical, and unverified autonomy poses risks. Without rigorous verification and validation, the deployment of autonomous AI in networks poses significant risks.

However, building a real-time, high-fidelity digital twin remains challenging. Unlike static systems, networks constantly evolve on fine timescales, driven by control-plane updates, traffic variability, failures, and operator interventions, making it exceptionally difficult to maintain a faithful, synchronized digital replica. Developing an effective twin requires detailed real-time telemetry between layers and devices, alongside tight integration of data collection, storage, and processing. On the emulation side, achieving meaningful fidelity entails more than simply replicating raw state; it demands accurate emulation of control logic, failure modes, and emergent behaviors under dynamic traffic loads and diverse fault conditions—demands often exceeding the capabilities of traditional simulators and offline models. Figure 1 illustrates the NDT concept: a programmable physical network is mirrored by a digital twin running on a SmartNIC, with a monitoring agent continuously synchronizing state from the physical to the digital environment.

In this work, we introduce GhostTwin, a lightweight, easy-to-deploy framework for defining and operating high-fidelity NDTs in a self-contained environment. To our knowledge, GhostTwin is the first open-source [6] Proof-of-Concept (PoC) deployment of a NDT. It enables creation of digital representations of programmable physical networks by emulating device operations—such as configuration and behavior—on a SmartNIC-based hardware emulator [9]. By leveraging hardware-level emulation, GhostTwin improves realism and scalability of digital twins, capturing physical effects like packet loss and residual bandwidth. Additionally, it supports various types and granularities of network data, making it adaptable to diverse performance and fidelity requirements.

2 MOTIVATION & CHALLENGES

Achieving a high-fidelity NDT requires addressing two fundamental challenges: data collection and digital representation. Prior work [5,

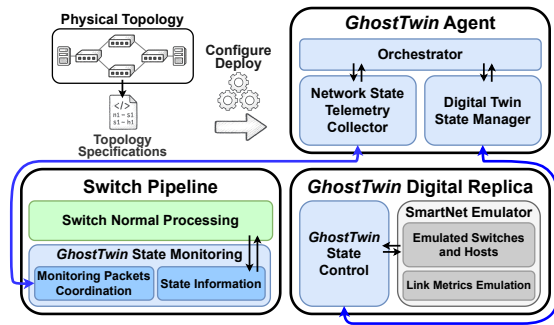


Figure 2: GhostTwin architecture.

7] primarily focused on the management and deployment of NTD models, with limited attention to the data acquisition methodologies and protocols.

Depending on the characteristics of physical infrastructures, network simulators such as ns-3, NS4, and PFPSim can model network elements. These simulators typically rely on discrete-event simulation, capturing system behavior as a sequence of time-ordered state changes, well-established in the literature. However, enabling live, high-fidelity NDT deployment demands advancing beyond these conventional simulation-based approaches. Next, we delve into these two challenges.

Collecting actual network state. A key challenge in building high-fidelity NDTs is acquiring comprehensive, real-time telemetry from physical infrastructure. Networks are distributed and composed of heterogeneous devices, each generating data at varying granularities, including flow-level statistics, control-plane updates, and queueing states. Conventional methods—like static configurations and export protocols such as NetFlow or sFlow—are limited in capturing the dynamic, fine-grained behaviors needed for accurate digital representations. These approaches often trade off overhead for fidelity: fine-grained telemetry (e.g., per-packet or per-flow) improves accuracy but adds significant network overhead, while coarse-grained sampling may miss critical transient events like microbursts or sub-RTT congestion. Achieving high-fidelity NDTs thus requires moving beyond traditional telemetry toward more adaptive, low-overhead, high-resolution data collection strategies. **High-fidelity twin representation.** Beyond raw state replication, an NDT must accurately emulate the behavior of the live network, including control logic, failure propagation, and traffic-load interactions. High-fidelity modeling requires capturing not just device configurations (e.g., forwarding rules, etc), but also implicit behaviors (e.g., queueing depths and delays). However, overly detailed models (e.g., packet-level software emulation) become computationally intractable at scale, in addition to not providing intrinsic hardware characteristics, which also reduces its fidelity.

3 GHOSTTWIN ARCHITECTURE & DEMO

Our proposed architecture is built on three core pillars: (i) programmable networks, (ii) SmartNICs, and (iii) a synchronization agent. Figure 2 shows the interaction between them.

Implementation. Our PoC implementation uses two programmable Tofino-based routers (EdgeCore Wedge 100-32X) as a physical testbed. These routers are programmed with a P4 code to forward IP network traffic. In addition, the exit pipeline is configured to

export telemetry at the desired granularity, allowing fine-grained data extraction. To this end, we define a custom probing packet structure that, upon reception, triggers data collection stored in local registers. The P4 code specifies which data to store. For this PoC, we configured the system to collect residual bandwidth, although other metrics could also be gathered and stored internally.

We use a hardware-based SmartNIC network emulator based on P7 [4, 8] to replicate the physical network. This emulator ensures realistic and efficient behavior replication by operating close to the network layer. It can emulate the entire physical topology and accurately mimic the networking behavior observed in practice. To maintain synchronization between the physical and digital counterparts, our GhostTwin agent periodically transmits probing packets to the switches. Upon receiving responses, it updates the digital replica and ensures consistency across both environments.

Limitations. GhostTwin is designed for deployment in programmable network environments, where it can fully leverage the capabilities of modern hardware and software-defined infrastructure. Nevertheless, it can also be adapted to operate in legacy networks by relying on standard telemetry protocols, such as NetFlow, sFlow, and static configuration files. However, GhostTwin currently supports digital twin execution on a single SmartNIC, which imposes inherent scalability limitations. This typically restricts its use to a few dozen switches, depending on factors such as network complexity, traffic volume, and overall emulation load.

During the demo. In the GhostTwin demo, attendees will observe how a programmable network can be mirrored into a digital twin using SmartNIC-based emulation. The demo will showcase how GhostTwin replicates device behavior and link dynamics, such as bandwidth, latency, and packet loss, with high fidelity. Attendees will see real-time traffic replay and metric visualization, demonstrating GhostTwin’s ability to reflect realistic network conditions and validate SmartNIC-based processing under controlled scenarios.

4 CONCLUSIONS AND FUTURE WORK

This demo presented GhostTwin, the first open-source PoC deployment of a network digital twin in programmable networks. Although our PoC demonstrates the practical feasibility of operating a Network Digital Twin in practice, significant challenges remain before its application in large-scale networks becomes viable. As future work, we plan to address the scalability of synchronization in Network Digital Twins, as well as gradually overcome current limitations in emulation fidelity.

ACKNOWLEDGMENTS

This work was supported by Ericsson Telecomunicações Ltda. , and by the Sao Paulo Research Foundation (FAPESP) , grant 2021/00199-8 (CPE SMARTNESS) , 2024/16207-8, 2020/05183-0 and 2023/00794-9. Also, it was supported by Foundation for Research of the State of Rio Grande do Sul (24/2551-0001394-6). This study was also partially funded by CAPES, Brazil-Finance Code 001, and CNPq (404027/2021-0). This work has been partially supported by the European Union’s Horizon Europe project under grant agreement No. 101070473 (FLUIDOS). Finally, the work was also funded by the Dutch 6G flagship project “Future Network Services”.

REFERENCES

- [1] Quazi Mishkatul Alam, Vinay Kolar, and Marina Thottan. 2025. Towards AI/ML-Driven Network Traffic Engineering. In *Proceedings of the 4th International Conference on AI-ML Systems (AIMLSystems '24)*. Association for Computing Machinery, New York, NY, USA, Article 24, 8 pages. <https://doi.org/10.1145/3703412.3703436>
- [2] Paul Almasan, Miquel Ferriol-Galmes, Jordi Paillisse, Jose Suarez-Varela, Diego Perino, Diego Lopez, Antonio Agustin Pastor Perales, Paul Harvey, Laurent Ciavaglia, Leon Wong, Vishnu Ram, Shihan Xiao, Xiang Shi, Xiang Cheng, Albert Cabellos-Aparicio, and Pere Barlet-Ros. 2022. Network Digital Twin: Context, Enabling Technologies, and Opportunities. *IEEE Communications Magazine* 60, 11 (Nov. 2022), 22–27. <https://doi.org/10.1109/mcom.001.2200012>
- [3] Daniele Brighenti, Guido Marchetto, Riccardo Sisto, and Fulvio Valenza. 2023. Automation for Network Security Configuration: State of the Art and Research Trends. *ACM Comput. Surv.* 56, 3, Article 57 (Oct. 2023), 37 pages. <https://doi.org/10.1145/3616401>
- [4] Fabricio E Rodriguez Cesen, Francisco Germano Vogt, Ariel Goes De Castro, and Christian Esteve Rothenberg. 2023. Towards multiple pipelines network emulation with p7. In *2023 IEEE 9th International Conference on Network Softwarization (NetSoft)*. IEEE, 290–292.
- [5] Hanshu Hong, Qin Wu, Feng Dong, Wei Song, Ronghua Sun, Tao Han, Cheng Zhou, and Hongwei Yang. 2021. NetGraph: An Intelligent Operated Digital Twin Platform for Data Center Networks. In *Proceedings of the ACM SIGCOMM 2021 Workshop on Network-Application Integration (Virtual Event, USA) (NAI'21)*. Association for Computing Machinery, New York, NY, USA, 26–32. <https://doi.org/10.1145/3472727.3472802>
- [6] INTRIG-Unicamp. 2025. GhostTwin. <https://github.com/intrig-unicamp/GhostTwin>. Accessed: 2025-07-15.
- [7] Marcelo C. Luizelli, Francisco G. Vogt, Paulo Silas Severo De Souza, Arthur F. Lorenzon, Roberto I. T. Da Costa Filho, Fabio D. Rossi, Rodrigo N. Calheiros, and Christian Esteve Rothenberg. 2024. DigiNet: Scaling up Provisioning of Network Digital Twin. In *2024 IEEE 10th International Conference on Network Softwarization (NetSoft)*. 136–144. <https://doi.org/10.1109/NetSoft60951.2024.10588932>
- [8] Fabricio Rodriguez, Francisco Germano Vogt, Ariel Góes De Castro, Marcos Felipe Schwarz, and Christian Rothenberg. 2022. P4 programmable patch panel (p7) an instant 100g emulated network on your tofino-based pizza box. In *Proceedings of the SIGCOMM'22 Poster and Demo Sessions*. 4–6.
- [9] Francisco Germano Vogt, Christian Rothenberg, Victor H. S. Lopes, Marcelo Caggiani Luizelli, Fabricio Rodriguez, Chrysa Papagianni, and Gergely Pongrácz. 2025. DEMO: SmartNet: Bridging Performance and Realism in Network Emulation with SmartNICs. In *ACM SIGCOMM 2025 Conference (SIGCOMM Posters and Demos '25)* (Coimbra, Portugal). ACM, Coimbra, Portugal, 3. <https://doi.org/10.1145/3744969.3748449>